

MENTIS · Procurement Intelligence Brief

Substance: 0.9% saline

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Executive Summary

0.9% saline, or normal saline, is an isotonic intravenous fluid primarily used for volume resuscitation, electrolyte replenishment, and as a carrier for drug delivery in clinical settings. The market reflects steady growth, projected to expand at a 7.4% CAGR through 2030, driven by rising surgical volumes and chronic disease prevalence [Tavily](#). Regulatory oversight plays a critical role in ensuring supply stability; facilities like Baxter's Spain plant and Fresenius Kabi's Norway site meet stringent cGMP standards [FDA](#). Key suppliers include Baxter, Fresenius Kabi, and B. Braun, with production hubs addressing global demand and prior shortages [Grand View Research](#). Commodity pricing pressures persist, with bulk configurations offering cost efficiencies, though risks such as hyponatremia and supply disruptions pose considerations for procurement strategies [Pipeline Medical](#).

Product Profile

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Clinical Use & Mechanism

0.9% saline, commonly referred to as normal saline, serves as a cornerstone in clinical settings due to its isotonic properties. Its primary applications include volume resuscitation, electrolyte replenishment, and as a vehicle for drug delivery. Beyond these standard uses, the solution has been explored in diverse surgical and perioperative contexts to optimize patient outcomes.

For instance, recent studies have considered its role in conjunction with agents like tranexamic acid for managing blood loss in orthopedic surgeries, particularly in elderly patients undergoing procedures such as those for peritrochanteric fractures. Although intravenous administration of normal saline with tranexamic acid has been recommended, the optimal dosing and administration routes remain subjects of clinical debate [source](#).

Additionally, normal saline supports the localized application of other agents, such as hemocoagulants, during surgical interventions like primary total hip arthroplasty (THA). One study highlighted its utility as part of a delivery mechanism for hemostatic agents such as 10 units of hemocoagulase bothrops atrox to effectively manage perioperative bleeding [source](#).

Although its traditional uses are well documented, emerging applications—such as ozonated saline for alternative therapeutic effects—remain in early stages of research, necessitating further evaluation to determine clinical relevance [source](#). This evolving evidence underscores both the versatility and the continuous innovation surrounding 0.9% saline in medical practice.

Market Size & Demand Drivers

The market demand for 0.9% saline, commonly used in intravenous (IV) therapy, wound care, and medication dilution, is substantial and steadily growing. Globally, the physiological saline market was valued at USD 6.73 billion in 2025 and is projected to expand at a compound annual growth rate (CAGR) of 7.40%, reaching USD 9.99 billion by 2030. This is driven by rising surgical volumes and the increasing burden of chronic diseases and trauma cases, where sterile 0.9% saline is critical for fluid replacement and wound irrigation in hospital settings [Tavily Tavily](#).

The demand is further fueled by strong growth in injectable and IV solution applications, which play a pivotal role in emergency care and routine clinical therapies. For example, the injectables/IV solutions segment is projected to experience the fastest growth, due to the widespread preference for 0.9% saline as a resuscitation and hydration agent in acute care units [Tavily](#).

However, the market also faces significant challenges. Pricing pressure is acute, as 0.9% saline is perceived as a commodity product. With numerous manufacturers globally producing similar solutions, intense competition has driven profit margins downward, constraining profitability [Tavily Tavily](#). Despite its robust growth trajectory, these dynamics could limit financial upside for suppliers focusing on standard formulations without differentiation.

Top Manufacturers & Suppliers

The production of 0.9% saline, also known as normal saline, is supported by a network of global manufacturers, many of whom play critical roles in ensuring its availability for healthcare and research needs. Key players in this field include Baxter, Fresenius Kabi, B Braun Medical, and Thermo Fisher Scientific.

Baxter International operates a significant manufacturing facility in Sabinanigo, Spain, which was approved by the U.S. FDA to produce 0.9% Sodium Chloride Injection, USP, in presentations of 250 mL, 500 mL, and larger volumes. This approval enhances supply stability and addresses historical shortages of saline solutions in the U.S. market, enabling better response to healthcare demands ([source](#), [source](#)).

In Europe, Fresenius Kabi Norge AS manufactures 0.9% Sodium Chloride Injection at its facility near Halden, Norway, which complies with FDA standards for good manufacturing practices ([source](#)). Meanwhile, B Braun Medical specializes in producing a sterile isotonic saline solution for irrigation, leveraging its expertise to meet a range of medical and surgical needs ([source](#)).

Globally, markets like China are dominated by manufacturers such as the Kelun Group. With its vast domestic manufacturing capacity, the company leads the sector by addressing growing local healthcare demands ([source](#)).

As demand for 0.9% saline continues to rise due to its critical role in medical care, the focus on optimizing production and distribution remains imperative for sustaining global health systems ([source](#)).

Regulatory & Compliance

The regulatory landscape for 0.9% Sodium Chloride Injection (Normal Saline) reflects the FDA's continued focus on ensuring safety, supply reliability, and compliance with manufacturing standards. Fresenius Kabi's facility near Halden, Norway, consistently adheres to [current Good Manufacturing Practices \(cGMP\)](#), as confirmed during its latest FDA inspection, supporting its authorized role in U.S. saline production and importation.

In response to disruptions, the FDA has recently approved Baxter's Sabinanigo, Spain, plant as a key manufacturing site for 0.9% Sodium Chloride Injection, USP, which bolsters the supply chain and enhances product availability across multiple packaging sizes (250 mL, 500 mL, and 1,000 mL). This regulatory milestone aims to address healthcare providers' previous challenges in maintaining access to saline solutions, particularly during shortages exacerbated by natural disasters and pandemic-related demands [like past shortages affecting the saline market](#).

Although the FDA officially resolved the shortage on August 8, 2025, nuances persist. Production levels have returned to pre-hurricane volumes, but stakeholders emphasize that resilience in this critical supply chain is an ongoing focus, requiring continuous vigilance and capacity-building measures [to ensure reliability](#). Furthermore, isolated events, such as B. Braun's voluntary [nationwide recall](#) of defective lots in 2024, underscore the necessity of stringent QA protocols and quick responses to protect public health.

Overall, while significant progress has been made to stabilize supply, regulators and manufacturers must remain proactive in addressing ongoing challenges, ensuring compliance, and anticipating future disruptions in this essential market.

Sourcing & Pricing Channels

The sourcing and pricing dynamics of 0.9% sodium chloride (normal saline) reveal a market characterized by scale-driven cost efficiencies and category-specific variability. Normal saline is widely used in healthcare for purposes such as fluid resuscitation, medication dilution, and hydration therapy [Pipeline Medical](#).

Wholesale prices vary depending on packaging format, with bulk configuration offering cost advantages. For instance, a 20-pack of 500 mL saline bags from Nextgen Pharmaceuticals is listed at \$64.17 [DrugPatentWatch](#). This translates to approximately \$3.21 per unit, making it an economical option for high-volume medical settings. By contrast, single-use vials, such as a 10 mL sodium chloride injection available in a 25-pack for \$34.22, are priced higher per mL to ensure sterility and precision in smaller medical applications [Westend Medical Supply](#).

The global physiological saline market is expected to experience sustained demand due to its essential role in emergency medical care and routine healthcare services. Forecasts indicate moderate growth through 2030, driven in part by increasing healthcare penetration and broader hospital usage [Research and Markets](#).

However, periodic supply disruptions, such as IV fluid shortages, may influence stockpiling behaviors and procurement strategies [ASHP](#). Healthcare providers often adjust policies to minimize waste, such as completing an existing infusion bag before switching to an alternative solution. This highlights operational complexities in balancing economic efficiency with clinical requirements in sourcing decisions.

Risks & Alternatives

0.9% Sodium Chloride (Normal Saline) is a widely used intravenous fluid, valued for its isotonic properties with an osmolality of 308 mOsm/L and a specific composition of sodium (154 mEq/L) and chloride (154 mEq/L) [6](#). However, its application is not without risks. Administering more than 1L per day may exceed physiological sodium and chloride levels, potentially causing **hypernatremia** and **hyperchloremic metabolic acidosis** [2](#). Furthermore, rapid correction of hypernatremia in adapted patients can lead to cerebral edema, posing a critical safety concern [1](#). There is also emerging evidence suggesting links between normal saline use and increased risks of kidney injury and metabolic acidosis due to its relatively high chloride concentration compared to "balanced" alternatives like Ringer's Lactate or Plasma-Lyte A [4](#).

To mitigate these risks, healthcare providers are considering alternatives. Balanced crystalloid solutions, which closely mimic physiological conditions, have shown promise in reducing metabolic disruptions while maintaining efficacy [5](#). While normal saline remains a cornerstone of intravenous therapy, its side effects must be carefully weighed against its intended clinical outcomes, especially in high-risk populations such as patients with congestive heart failure or severe renal insufficiency [3](#).

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